# VORLD X ATTENDANCE 360

CIN: U62099UP2026PTC245598

## Executive Project Report & Technical Specification

Enterprise-Grade Biometric QR Terminal, Native Android APK & Hybrid Attendance Management System

| ORGANIZATION | LOCATION | SCOPE DELIVERED | RELEASE DATE |
|---|---|---|---|
| VorldX Industries Pvt Ltd | Lucknow HQ, Uttar Pradesh | 18 Screens • RBAC • Mobile APK | October 2026 (Production v2.4) |

## 1 Project Overview & Core Architecture

The **VorldX Employee Attendance System** is an end-to-end enterprise attendance infrastructure built for VorldX Industries Private Limited. The system completely modernizes physical fingerprint scanners and paper logbooks using rotating cryptographic QR biometrics, automated late/grace policy validation, multi-tier Work From Home (WFH) & Leave workflows, and dedicated native Android APK distribution.

### 🏢 HR & Admin Portal

Full web dashboard with Super Admin & HR Manager roles. Real-time KPIs, staff directories, attendance regularizations, policy rules, and multi-display pairing.

### 📱 Employee Mobile APK

Native Android APK (Capacitor) featuring biometric QR camera scanning, multi-day calendar range pickers for Leaves & WFH, and interactive monthly attendance history.

### 📺 Office Kiosk Display

Wall-mounted full-screen kiosk interface displaying auto-rotating QR tokens (45s validity), animated countdown progress, and live check-in confirmation chime.

## 2 Technology Architecture & Data Schemas

### 💻 Technical Stack

**Frontend Web:** React 19, Vite 8, Vanilla CSS Design System, Lucide Icons

**Mobile APK:** Capacitor Native Android (Java 21, Gradle, MinSDK 22)

**Backend API:** Node.js, Express 5 REST API (Port 5000)

**Database:** SQLite with WAL Mode (Zero-latency offline file storage)

**Security:** Cryptographic Rotating Tokens (45s TTL), RBAC Authorization

### 🗄️ Relational Data Models

`employees`: ID (VX001-VX050), Name, Role, Shift, Dept, Avatar, Phone

`attendance`: Date, Check-in, Check-out, Duration, Status, Kiosk ID

`wfh_requests`: Date Range, Reason, Approval Status, Reviewer Note

`leave_requests`: From/To Date, Type (Casual/Sick), Reason, Status

`kiosk_displays`: Registry ID, Location, Token, Active Status

## 3 Hierarchical WFH & Leave Approval Workflow

```
[Employee Submits Multi-Day WFH / Leave] │ ▼ [HR Manager Queue: Sneha Verma] ──(Approves / Rejects)──▶ [Employee Sees Live Status & Remarks] └──▶ [Auto-updates Attendance Calendar] [HR Manager Submits Own WFH Request] │ ▼ [Super Admin Queue: Vikram Singhania] ──(Approves/Rejects)──▶ [HR Sees Decision & Logged in Audit]
```

## 4 Catalog of All 18 User Interface Screens Implemented

| Screen | Screen Title | Platform | Key Implementation Features | Status |
|---|---|---|---|---|
| **Screen 1** | Admin & HR Login | WEB PORTAL | Role-aware authentication for Super Admin & HR Manager with 1-click switcher. | VERIFIED |
| **Screen 2** | Admin Executive Dashboard | WEB PORTAL | Month picker (Oct 2026), 4 KPI cards (Total, Present, WFH, Late) & daily bar chart. | VERIFIED |
| **Screen 3** | Employee Directory | WEB PORTAL | Department filter, status badges, staff table pagination & + Add Employee modal. | VERIFIED |
| **Screen 4** | Employee Profile (Admin) | WEB PORTAL | Breadcrumbs, avatar, VX001 badges, tabs: Overview, Attendance, WFH, Personal. | VERIFIED |
| **Screen 5** | Attendance Management | WEB PORTAL | Date range filter, CSV export, manual regularization & audit notes. | VERIFIED |
| **Screen 6** | WFH & Leave Approvals | WEB PORTAL | Queue routing (HR vs Admin), status tabs (All, Pending, Approved), 1-click decision. | VERIFIED |
| **Screen 7** | Policy & Shift Settings | WEB PORTAL | General shift (09:00 AM - 06:00 PM), 15m grace period, late rules & workdays. | VERIFIED |
| **Screen 8** | Office Kiosk Displays | WEB PORTAL | Registry of 4 displays (Reception, Floor 2, Floor 3, Meeting Room), pairing tokens. | VERIFIED |
| **Screen 9** | Reports & Graphical Analysis | WEB PORTAL | 85% Attendance rate, daily attendance bar chart & department distribution donut chart. | VERIFIED |
| **Screen 10** | Compliance Audit Trail | WEB PORTAL | Immutable activity logs tracking actions, timestamps, and actors. | VERIFIED |
| **Screen 11** | Mobile App Authentication | ANDROID APK | Employee ID login (VX001), corporate branding, fast authorization. | VERIFIED |
| **Screen 12** | 6-Box OTP Verification | ANDROID APK | Auto-focused 6-digit OTP inputs, 30s countdown timer, resend action. | VERIFIED |
| **Screen 13** | Mobile Home Dashboard | ANDROID APK | Live greetings, Today punch card (In: 09:05 AM, Out: 06:02 PM), Scan QR button. | VERIFIED |
| **Screen 14** | Camera QR Scanner | ANDROID APK | Real camera viewfinder with Html5Qrcode, target reticle, torch & audio chime. | VERIFIED |
| **Screen 15** | WFH & Leave Requests Hub | ANDROID APK | Calendar range pickers (From/To), auto day calculation, HR decision tracking. | VERIFIED |
| **Screen 16** | Attendance Calendar & Logs | ANDROID APK | Material calendar view, clickable DD/MM/YYYY dates, punch reasons & terminal ID. | VERIFIED |
| **Screen 17** | Employee Profile Self-Service | ANDROID APK | Self photo avatar presets, editable contact/email, password change & locked badges. | VERIFIED |
| **Screen 18** | Rotating Office QR Terminal | KIOSK TV | Fullscreen high-contrast display, live clock, 45s dynamic rotating QR biometric. | VERIFIED |

## 5 Role-Based Access Control (RBAC) Matrix

| Module / Permission | Super Admin | HR Manager | Employee |
|---|---|---|---|
| **Executive Dashboard** | Full corporate KPIs & bar charts | Department KPIs & summaries | Personal attendance overview |
| **Employee Records** | Full CRUD & credential management | Staff records & shift assignments | View & edit own profile |
| **Attendance Logs & Edits** | All records & manual regularization | Staff regularizations with audit note | View personal logs & reasons |
| **Staff WFH / Leave Approval** | Override authority if needed | Primary Reviewer & Approver | Submit applications & track review |
| **HR Lead WFH Approval** | Primary Reviewer & Approver | Submits own requests to Admin | No access |
| **Shift Policies & Grace** | Full management & thresholds | Read-only inspection | No access |
| **Kiosk Display Pairing** | Add/Deactivate/Re-token displays | View display online status | Scans kiosk with phone camera |
| **Audit Trail & Reports** | Full immutable compliance export | Department analytical reports | View personal attendance report |

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**18 / 18 SCREENS VERIFIED**

Official Internship Project Sign-off